

(SHWASI) Smart Hand Wearable Aid for Speech Impaired: Sign Language Communication using Flex Sensor-based Finger Spelling

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Abstract—The paper presents a novel Smart Hand Wearable device for sign language recognition and communication. By strategically placing flex sensors on a glove, the system translates finger movements into binary data, enabling users to select from three operational modes: common phrases, English alphabets, and numbers/special characters. A voltage divider circuit ensures accurate sensor readings, while an operational amplifier further refines the data. Additionally, a Bluetooth module facilitates wireless communication with a smartphone application. Implemented modes allow users to convey messages seamlessly through intuitive finger gestures. The system's robustness and reliability are demonstrated through practical implementation and results. This innovative approach offers a cost-effective and user-friendly solution for speech-impaired individuals, significantly enhancing accessibility and communication. The key takeaway is the system's ability to provide an efficient and practical means of communication for those with speech impairments.

Keywords — *Flex Sensor, Finger Spelling, Smart Hand-Wearable, Speech Impaired*

I. INTRODUCTION

Sign language recognition is being increasingly recognized as a vital research area, focused on enhancing communication for individuals with hearing and speech impairments by translating gestures into text or speech [1]. Finger spelling, a modified version of sign language with alphabets, numerals, special characters, and default statements, is another useful tool for communication [2]. Two primary methods employed for sign gesture detection are: image-based which involves collection of images of hand signs [3], [4], and sensor-based systems where sensors are used for capturing data [5], [6], [7]. There are several methods for enhancing the system accuracy [8], [9]. Although image-based systems can be effective, they often face challenges such as lighting variations and background interference [3]. Conversely, sensor-based systems are more suitable for real-time applications but may be cumbersome and uncomfortable to wear [10].

Recent works in sensor-based systems have explored various methods [3], including surface electromyography (sEMG) to capture muscle tissue activity [11], [12], [13], and combinations of flex and inertial sensors [14], [15], [16] or use of only inertial sensor [17], [18]. Custom sensors [3], such as those coated with Reduced Graphene Oxide (RGO) [19] and Pyrolytic Graphite Sheets (PGSSs) [20], have improved flexibility and sensitivity. A gesture recognition system using four flex sensors, speakers, a

Raspberry Pi 3B+, was introduced, with each sensor corresponding to a distinct message [21].

A model for recognizing Indian Sign Language gestures using flex sensors and an Inertial Measurement Unit is proposed [22]. The IMU tracks hand movement, and flex sensors monitor finger orientation. Like vision-based approach, sensor-based methods offer numerous possibilities to enhance system accuracy and performance [23]. However, adding more sensors for performance enhancement of the system escalates costs and space requirements. To address this, a cost-effective approach using five flex sensors on one hand is suggested, creating five binary units, allowing for 32 unique combinations. Each alphabet is assigned a specific combination, avoiding the need for additional sensors, thus reducing production costs.

Another work introduced two modes: one for wheelchair movement governance using flex sensor and another combining a flex sensor with a voice module for audio output [24]. A smart glove capable of converting hand gestures into text was proposed, also functioning as an obstacle detection system [25]. Another work [26] is upgraded with multiple modes of operation for a seamless user experience. The resistance variation in the flex sensor [27] corresponds to the degree of finger bend, resulting in distinct values when the finger is bent or straight. These values are processed by the Arduino UNO and displayed using an LCD output [18], or interfacing with a computer [28], or an application developed using MIT App inventor, aiding in audio-visual output [29].

II. METHODOLOGY

A. Working

The Smart Hand Wearable Device, abbreviated as SmWaD, represents a ground-breaking advancement in the realm of gesture-based communication technologies for speech impaired individuals. By employing flex sensors strategically positioned on each finger of a wearable glove made up of neoprene and polyester fabric, this system harnesses the subtle movements of the user's hand to convey binary data, effectively transforming the glove into a multifaceted input device. Through the nuanced interplay of finger positions—where in a straightened finger signifies a binary 0 and a bent finger indicates a binary 1—the glove can encode a diverse array of binary combinations. These combinations, in turn, serve as the digital language through which the user interacts with the device.

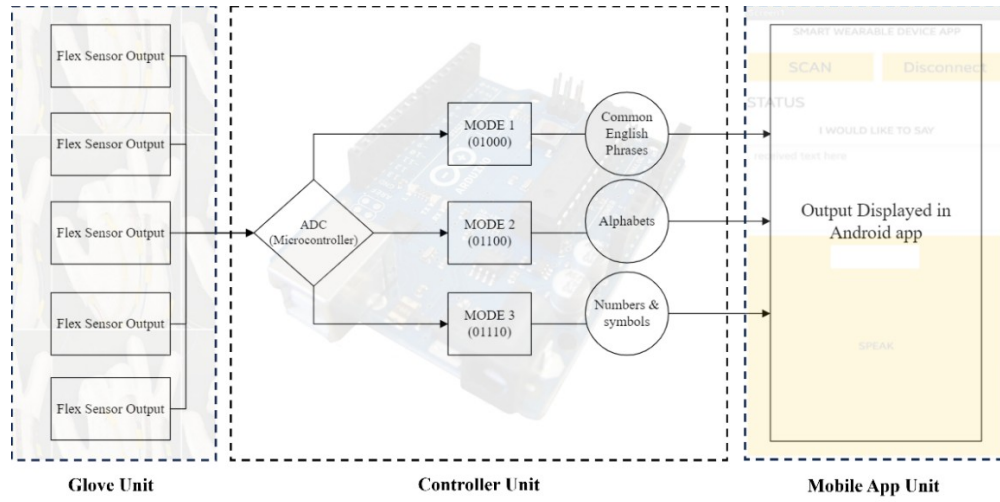


Fig. 1. Schematic flow diagram of working of Smart Hand-Wearable

Central to the functionality of the Smart Hand Wearable are its three distinct modes, each tailored to accommodate different communication needs. Mode 1 caters to the expression of common English phrases, while Mode 2 facilitates the selection of individual English alphabets, and Mode 3 enables the input of numbers and special characters. The activation of these modes is elegantly orchestrated through specific finger configurations, with each mode associated with a unique binary code. By executing the prescribed finger movements, users seamlessly transition between modes, thereby accessing the desired functionality with ease and efficiency. Upon activation, the user navigates through the available options within the selected mode by manipulating their fingers to generate the corresponding binary combinations. These combinations serve as the digital keys that unlock a vast repertoire of pre-defined messages, meticulously catalogued in Table I. As users articulate their intentions through the intricate choreography of finger gestures, the system's embedded microcontroller swiftly processes these inputs, decoding the binary signals and translating them into actionable commands. The culmination of this process occurs with the transmission of the selected message to a paired smart device via an integrated Bluetooth module. This communication bridge seamlessly integrates the capabilities of the Smart Hand Wearable with the expansive functionality of modern mobile technology, enabling users to effortlessly convey their thoughts, commands, and inquiries through intuitive hand movement.

B. Circuit Design

A flex sensor is a type of adjustable resistive element in which the amount of deflection it experiences is exactly proportional to its electrical resistance. To enhance precision in measuring and sending flex sensor data to the Arduino Uno microcontroller, a voltage divider circuit as shown in Fig. 2, combined with an operational amplifier (LM324), acting as an impedance buffer was designed. Equation (1) gives the voltage developed across R_1 resistor in the voltage divider circuit.

$$V_{out} = V_{CC} \frac{R_1}{R_f + R_1} \quad (1)$$

R_f refers to the resistance provided by the flex sensor, R_1 is the resistance of the resistor connected in series to the flex sensor, V_{CC} is the voltage supply provided to the flex sensor. The

operational amplifier serves as an intermediary between the low-impedance flex sensor and the high-input impedance ADC ports of the Arduino Uno. This resolves the impedance mismatch, ensuring minimal signal attenuation and reducing susceptibility to noise. The unity voltage gain of the buffer preserves the integrity of the sensor signal, enabling accurate readings by Arduino Uno. The operational amplifier serves as an intermediary between the low-impedance flex sensor and the high-input impedance ADC ports of the Arduino Uno. This resolves the impedance mismatch, ensuring minimal signal attenuation and reducing susceptibility to noise. The unity voltage gain of the buffer preserves the integrity of the sensor signal, enabling accurate readings by Arduino Uno. By implementing this circuit design, we ensure robust and reliable performance of the flex sensor system, making it suitable for our application requiring accurate measurement and wireless communication capabilities.

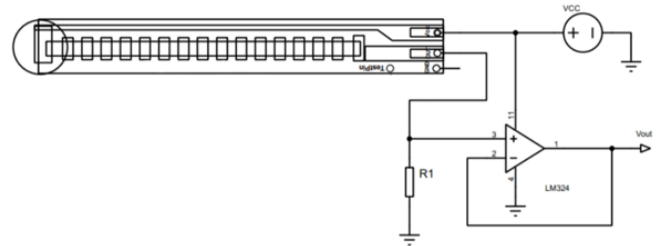


Fig. 1. Circuit schematic of Flex Sensor connection

C. Android Application

Furthermore, to enable wireless communication between the Arduino Uno and a smartphone, a separate Bluetooth module (HC-05) is integrated into the circuit. The application, developed using the MIT App Inventor platform, integrates three fundamental modules: Bluetooth functionality, a text box, and a text-to-speech converter. Upon establishing a connection with the HC-05 Bluetooth module, serial data is received and stored. This data serves as the basis for populating the text box and facilitating the conversion of text to speech. Notably, the application provides users with functionalities such as scanning for available Bluetooth devices, establishing, or terminating

connections, displaying connection status, presenting text, and audibly articulating the text contents of the text box. The front-end display is shown in Fig. 3.

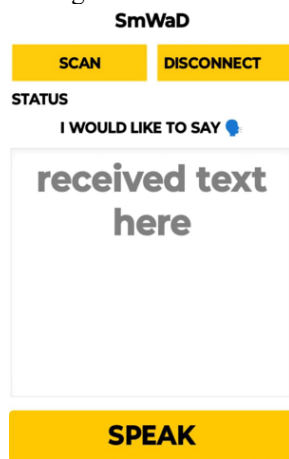


Fig. 2 Application Front-End Design

III. HARDWARE AND SOFTWARE COMPONENTS

Below listed are the components which are utilized in development of the Smart Hand-Wearable technology.

1. Flex Sensors

Flex sensors are flexible devices designed to detect and measure bending. They consist of thin, conductive material printed on a flexible substrate. Two 4.5 inches and three 2.2 inches flex sensor are used for measuring the finger flexing intensity. Any change in the position of the fingers is reflected out as change in the resistance value of the flex sensors.

2. LM324 Operational Amplifier

The LM324 is a quad operational amplifier (op-amp) integrated circuit. It contains four independent op-amps that can be operated from a single voltage supply. Employing the opamp helps in balancing the impedance mismatch, minimize signal loss and achieve better noise immunity.

3. HC-05 Bluetooth Module

The HC-05 Bluetooth module is a widely used device for wireless communication with Arduino. Operating on Bluetooth 2.0 + EDR technology, it supports a range of about 10 meters. The module operates in the 2.4 GHz ISM frequency band. It is employed to communicate the textual phrases generated by Arduino Uno to the Android App.

4. Arduino Uno

The Arduino Uno is a versatile microcontroller board based on the ATmega328P chip. The microcontroller was employed to process the input from the flex sensors obtained via the ADC port and provide the corresponding output to the Android App via the Bluetooth module.

5. MIT App Inventor

MIT App Inventor is a web-based integrated development environment (IDE) that allows users to create mobile applications for Android devices. A user-friendly application interface is developed using the IDE which can speak out the phrases and provide a visual representation of the same.

Table II gives out the technical specifications of the hardware components used in building the Hand-Wearable.

TABLE I FINGER SPELLING LOOKUP DIRECTORY

Combination	MODE 1	MODE 2	MODE 3
00000	Stop	Stop	Stop
00001	Good Morning	A	0
00010	Good Afternoon	B	1
00011	Good Evening	C	2
00100	Good Night	D	3
00101	Hi! I am	E	4
00110	What's your name?	F	5
00111	I am from	G	6
01000	MODE 1		
01001	Nice to meet you	H	7
01010	What do you do?	I	8
01011	I am	J	9
01100		MODE 2	
01101	Can you come again	K	!
01110			MODE 3
01111	Start	Start	Start
10000	Please	L	#
10001	I am sorry	M	\$
10010	Thank You	N	%
10011	Excuse me	O	^
10100	What do you think?	P	&
10101	That sounds great	Q	*
10110	That smells good	R	(
10111	Never mind	S	)
11000	I don't understand	T	-
11001	What do you mean?	U	
11010	Can I help you	V	+
11011	How much does it cost	W	=
11100	I will be with you in a moment	X	?
11101	Please call me back	Y	/
11110	My phone number is	Z	>
11111	Have a good day		<

TABLE II TECHNICAL SPECIFICATION OF HARDWARE COMPONENTS

S.No.	Component	Specification
1	Flex Sensor	Dimension: 2.2 inches, 4.5 inches Flat Resistance: 25kΩ (2.2 inches) 10kΩ (4.5 inches) Operating Voltage: 0 to 5V DC Operating Temperature: -45°C to +80°C Resistance Tolerance: ±30% Power Rating: 0.5W (continuous), 1W (peak)
2	Arduino Uno	Microcontroller: ATmega328P (8-bit AVR core) Clock Speed: 16 MHz Memory: Flash Memory: 32 KB SRAM: 2 KB Operating Voltage: 5V Digital I/O Pins: 14 Analog Input Pins: 6
3	HC-05 Bluetooth Module	Bluetooth Version: 2.0 + EDR (Enhanced Data Rate) Range: Up to 10 meters Operating Voltage: 3.3V to 5V DC Communication Interface: UART Data Rate: Up to 2.1 Mbps Frequency: 2.4 GHz ISM band

IV. RESULTS AND DISCUSSION

In a workplace event or a social gathering, the smart glove can be used by the user to interact with the society without any complex. The smart glove is designed to be used as finger spelling standalone system which is easy for the user to carry it anywhere, anytime. The working of the glove based on the finger spelling lookup table as in Table 1 is explained in this session. The smart glove works three different modes such that the user can build their own phrases and communicate. The precautions have been taken to avoid over riding or mis-finger spelling by other modes when a particular mode is in use. Basic condition for activation of mode 1, 2, or 3 is done by providing the corresponding finger spelling (01000, 01100, 01110). Once when the mode is active, the communication between the sensor and the SmWaD begins by providing the start combination 01111, which is common for all the modes. As the communication starts, the finger spellings are converted to their corresponding equivalents. Till the stop combination 00000 is provided the other mode activation combinations are made inactive.

Following the override and mis-finger spelling prevention process outlined earlier, the user can activate Mode 1 by providing the finger spelling combination of 01000. For which, the index finger is bent, and rest of the fingers are set upright to achieve the binary combination “01000” as shown in Fig. 4a. When mode 1 is on, the glove operates and prevalent phrases such as "Hello", "Nice to meet you", "Excuse Me", and "Thank You" are enabled. The microcontroller determines the mode of operation as “MODE 1” and unlocks the phrases available in that mode. For instance, if someone greets “Good Morning”, the user can greet back “Good Morning” by bending the little finger and straightening out rest of the fingers which generates the combination “00001” and the phrase “Good Morning” is displayed and read out by the app as shown in Fig. 4b. If asked for the user’s name, under Mode 1, user can make the combination “00101” by bending middle and little fingers, and the phrase “Hi! I am” is displayed and read out by the app. But the username is not a part of predefined standard phrases, so the user needs to spell the name using combination of characters using Mode 2.

Further, user can enable Mode 2 of operation for accessing alphabets, by making the combination “01100”. Once mode 2 is activated, as shown in Fig. 5a, user can spell out the name “XYZ” by making the combinations “11100”, “11101”, “11110” which are then displayed and read out by the app in a sequential fashion as “X Y Z”. Fig. 5b. shows the output “X” displayed in the screen of the application. Mode 3 of operation gives the user access to numbers and special characters. It is enabled by setting index, middle and ring fingers completely bent to achieve the combination “01110” as shown in Fig. 6. To prevent the overlapping of the phrases(words) and alphabets, the user must exercise a small delay between two consecutive actions. In this work, a maximum delay of 2 second and 4 seconds have been introduced to insert a blank space between the words and dot correspondingly. The complete phrase or word that needs to be communicate is completely defined by providing the finger combination, the speak button in the SmWaD can be used to speak out the information. This helps the user to make audio-based communication. The gloves

performance was validated across all the modes and combinations. Test data was collected from a total of 100 volunteers of varying age and gender. A total of 3 modes (mode 1, 2, and 3) and 88 combination (30,29,29) has been tested for 100 repetition and the overall accuracy was found to be 79%. The drop-in accuracy could be due to the strain realized by the flex sensor for a long duration while being validated. The accuracy and performance of smart glove can be enhanced by using the AI pretrained systems.

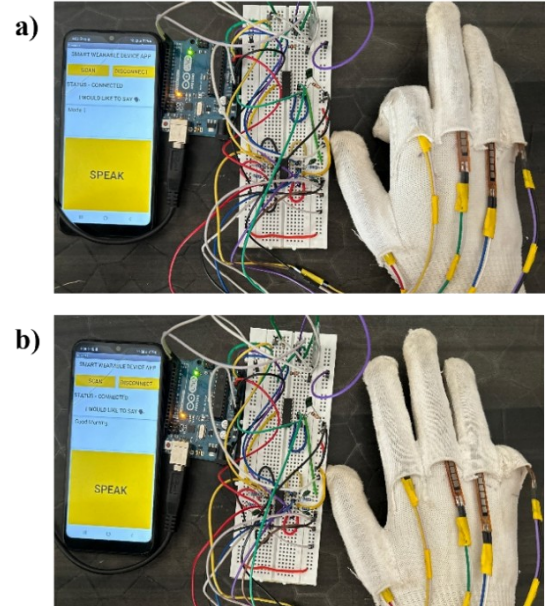


Fig. 3. (a) Activation of Mode 1 by providing the input combination 01000. (b) Displaying “Good Morning” message by providing the input combination 00001.

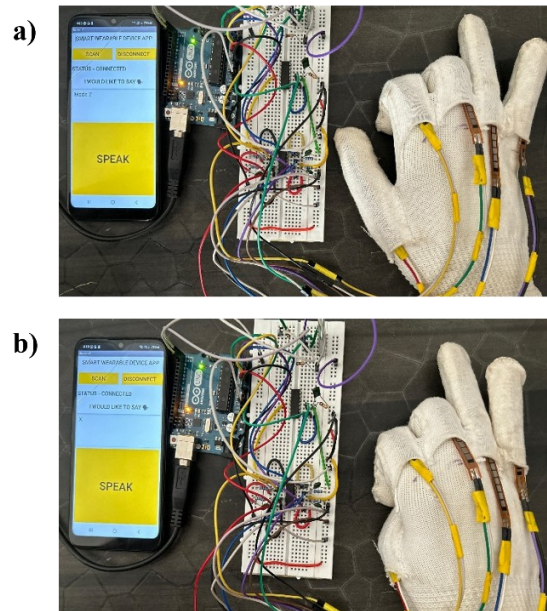


Fig. 4. (a) Activation of Mode 2 by providing the input combination 01100. (b) Displaying “X” message by providing the input combination 11100.

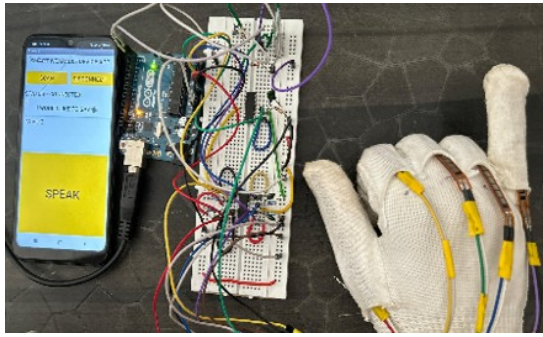


Fig. 5. Activation of Mode 3 by providing the input combination 01110.

V. CONCLUSION

The proposed methodology aims to facilitate communication between individuals with disabilities. The most popular approaches for detecting sign language are vision-based and sensor-based methods. Traditional sensor-based devices for communication often suffer from complexity, cost, or inefficiency, such as gloves with predefined sentences. This proposed system enhances communication possibilities by introducing three modes (Mode-1, Mode-2, and Mode-3) to a glove equipped with five flex sensors, simplifying design and reducing complexity by implementing finger spelling technique. By employing hybrid switching, users can communicate more easily, with the glove distinguishing hand gestures from predefined ones. This enhancement, alongside binary code allocation for gesture recognition, improves the glove's effectiveness and precision. Ultimately, our smart glove empowers individuals with disabilities to lead more independent lives, overcoming their challenges and fostering growth. While the proposed technique demonstrates efficiency, effectiveness, and precision, offering a promising solution for enhancing communication and autonomy for individuals with disabilities, further analysis of the test data indicated a decrease in device functionality as the user's age advances. This may be attributed to reduced hand dexterity, flexion, and strength. To accommodate variability and serve users of different age groups, we can utilize machine learning techniques. By training classification models on the collected test data, we can ensure consistent output for all gestures, irrespective of the user's age group.

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